Software Requirements Specification (SRS)

Functional Requirements

Functional requirements describe what the system must do. Each requirement has a unique ID, a priority and a way to verify that it works.

MoSCoW priorities: Must Have means essential, Should Have means important but not essential, Could Have means useful if time allows, and Won't Have means not planned for this version.

ID	Functional Requirement	Priority	Verification
FR-01	The system shall allow the Administrator to create, update and deactivate user accounts.	Must Have	Create, update and deactivate a test account and check the results.
FR-02	The system shall allow users to log in using a username and password.	Must Have	Test login with correct and incorrect credentials.
FR-03	The system shall restrict system functions according to the user's assigned role.	Must Have	Log in as each role and check that restricted functions are unavailable.
FR-04	The system shall allow Lecturers to create and manage courses.	Must Have	Create, update and view a test course using a Lecturer account.
FR-05	The system shall allow Lecturers to create assignments with descriptions and submission deadlines.	Must Have	Create an assignment and confirm that its details are saved.
FR-06	The system shall allow Students to view courses and assignments available to them.	Must Have	Log in as a Student and verify that only relevant courses and assignments appear.
FR-07	The system shall allow Students to submit assignments before the deadline.	Must Have	Submit an assignment before its deadline and confirm that the submission is recorded.
FR-08	The system shall allow Students to resubmit assignments before the deadline.	Must Have	Submit an assignment twice before the deadline and verify that the latest valid submission is identified.
FR-09	The system shall record late submissions and apply a penalty of 10% per day for up to three days.	Must Have	Test submissions one, two and three days late and check the applicable penalty.
FR-10	The system shall allow Lecturers to grant individual extensions to Students.	Must Have	Grant an extension to one Student and verify that the new deadline applies only to that Student.
FR-11	The system shall allow Lecturers to record marks and feedback for submitted assignments.	Must Have	Enter marks and feedback and confirm that they are saved.

FR-12	The system shall allow the HoD to review and moderate marks before their release.	Must Have	Submit marks for moderation and verify that the HoD can review them.
FR-13	The system shall prevent Students from viewing marks until the HoD has approved their release.	Must Have	Attempt to view unapproved marks and confirm that they remain unavailable.
FR-14	The system shall record changes made to marks, including who made each change and when it occurred.	Must Have	Change a mark and check that the change history contains the user and timestamp.
FR-15	The system shall allow Students to view released marks and feedback for their assignments.	Must Have	Release approved marks and verify that the relevant Student can view them.
FR-16	The system shall allow authorised users to view relevant assignment and assessment records.	Should Have	Log in as different roles and check access to the relevant records.

Non-Functional Requirements

Non-functional requirements describe how well the system must operate and the conditions it must satisfy. Each requirement below has a measurable target that can be tested.

ID	Category	Non-Functional Requirement	Priority	Verification
NFR-01	Performance	The system shall complete a login attempt within 2 seconds when the database contains records for 800 Students and their associated system data on the specified test computer.	Must Have	Measure login time with the test database populated.
NFR-02	Capacity	The system shall store records for at least 800 Students without losing or corrupting existing records.	Must Have	Populate the database with 800 Student records and verify that all can be retrieved correctly.
NFR-03	Security	The system shall store passwords as salted hashes and shall not store plain-text passwords.	Must Have	Inspect the database and verify that stored passwords are not plain text.
NFR-04	Access Control	The system shall prevent users from accessing functions outside their assigned roles in all defined access-control tests.	Must Have	Run access-control tests for all four roles and verify that unauthorised actions are rejected.

NFR-05	Reliability	The system shall correctly save and retrieve 100 consecutive test assessment records without losing or changing their stored values.	Must Have	Save and retrieve 100 test records and compare the results.
NFR-06	Data Integrity	The system shall record every successful mark change with the user ID, timestamp, previous mark and new mark.	Must Have	Change test marks and verify that all required details appear in the audit history.
NFR-07	Offline Operation	The system shall allow users to log in and access locally stored records without an internet connection.	Must Have	Disconnect the test computer from the internet and test login and record access.
NFR-08	Maintainability	The project shall build successfully using the supplied CMake configuration and the required C++17-compatible compiler.	Should Have	Configure and build the project on the specified development environment.